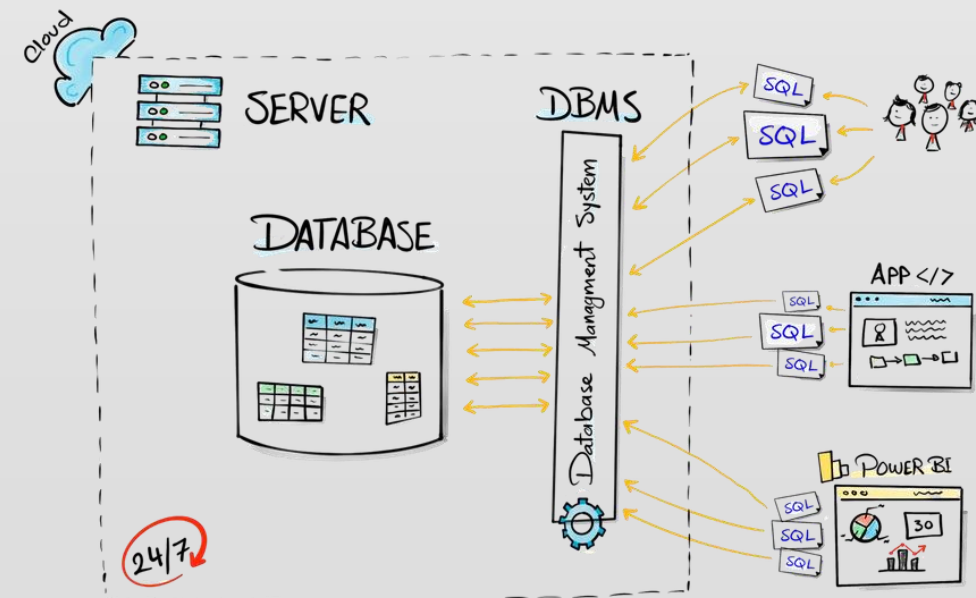




ICE ENTERTAINMENT

DIMENSIONAL MODEL DESIGN AND IMPLEMENTATION



EXECUTIVE SUMMARY

ICE Entertainment operates globally but lacks an integrated analytical platform due to fragmented Sales and ERP data

- A dimensional data model (star schema) is proposed to unify data and improve reporting
- Enables analysis of profitability across customers, products, stores, time, and markets
- Standardises financial data in USD for consistent global reporting
- Provides a scalable, data-driven foundation supported by data strategy canvas and data flow architecture





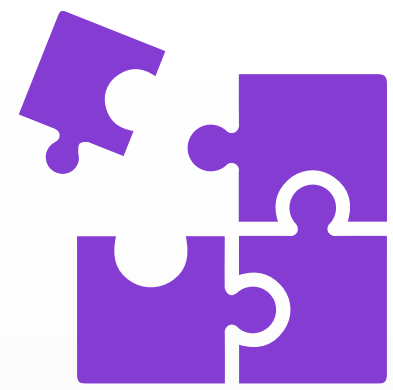
PROBLEM STATEMENT

ICE Entertainment runs on multiple systems that stores customer, product and transactional data separately.

At present, the company's management is unable to get the following insights from its database:



- ✓ Profitability and behavior of customers
- ✓ Performance of products and categories
- ✓ Efficiency of stores and regions
- ✓ Sales trends on the basis of time

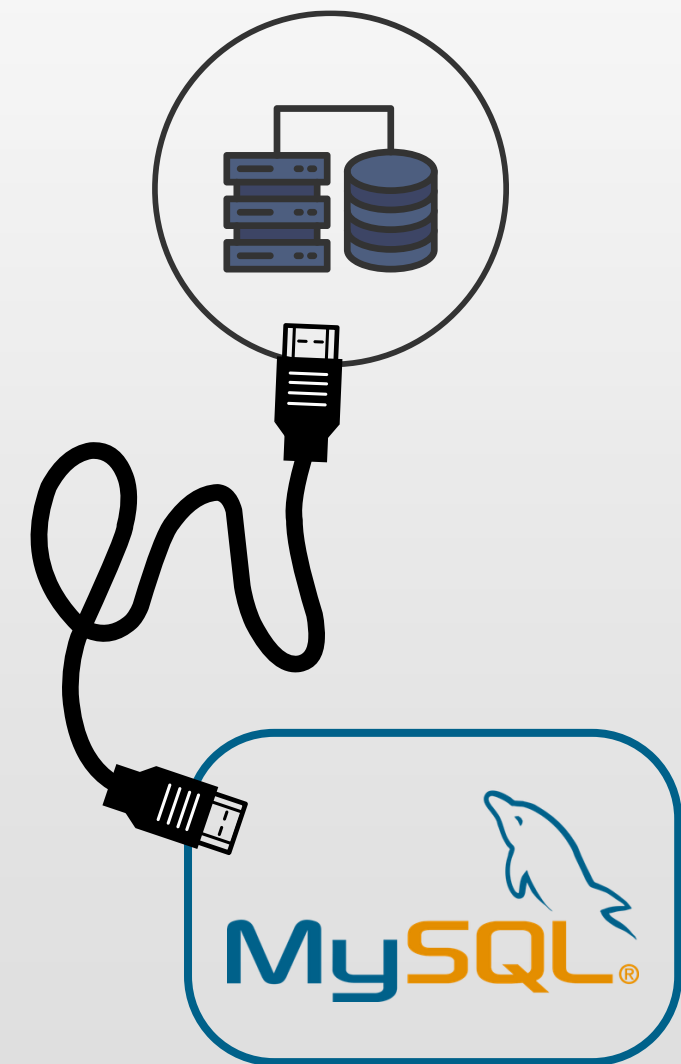


OBJECTIVE

To develop a dimensional data model that will allow data integration from different sources into one analytics platform.

This approach should result in:

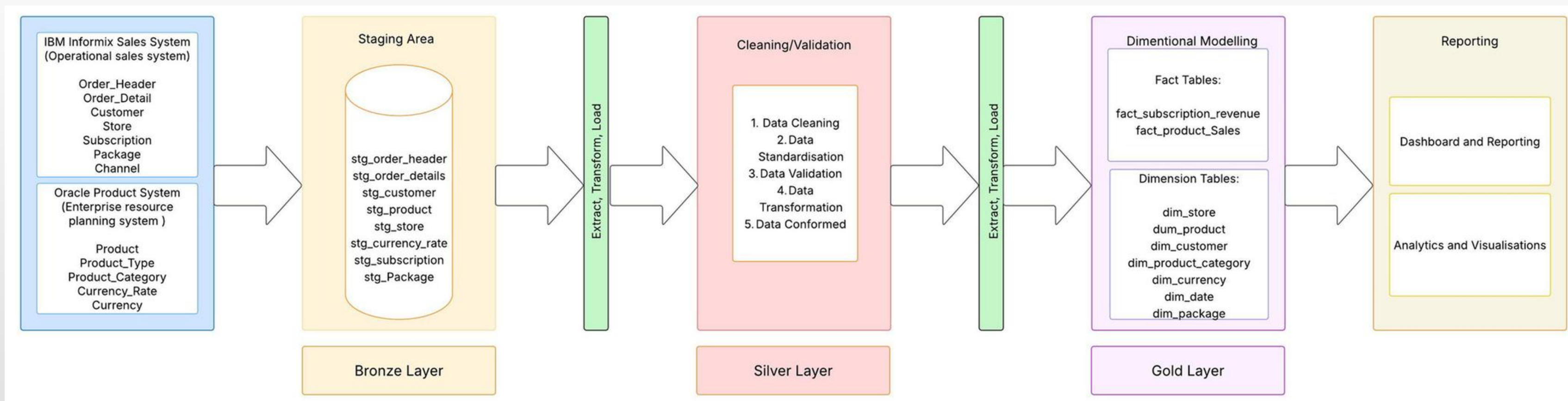
- A single source of truth for business reporting
- Analysis by various dimensions such as customers, products, stores, time, and market
- Decision-making based on effective data querying and analysis
- Useful information that will help to improve sales performance, customer targeting, and strategic planning



Strategic Objective	Strategic Framing		Strategic Enablement	
1. - Integrate Sales and ERP into a unified USD-based model - Create a single trusted global data source - Enable global sales visibility - Support data-driven decision-making	2. Key Decisions & Use cases: - Identify most profitable customers and products - Analyse store and market performance - Support sales and product decisions	3. AI / Analytics Opportunities: - Profitability analysis - Customer segmentation - Product performance insights - Predictive analytics (retention)	5. Data Requirements & Products: - Sales data + ERP data integration - Fact table (sales, cost, margin) - Key dimensions (customer, product, store, time)	6. Consumers & Users: - CEO (strategy) - Sales Manager (performance) - Purchasing Head (products) - CRM Team (customers)
	4. Value Measures & ROI / CBA - Faster and accurate reporting - Improved margin visibility - Better decision-making - Increased profitability		7. Technology & Systems: - Informix + Oracle (sources) - Pentaho (ETL) - MySQL (data warehouse)	
Strategic Readiness				
8. Portfolio Management - Enterprise analytics initiative - Supports multiple business functions - Enables future analytics capabilities	9. Operating Model & Governance - Standard definitions (sales, cost, margin) - USD currency standardisation - ETL data validation		10. Risks & Ethics - Data inconsistency - Currency conversion errors - Data quality issues - Privacy and security risks	



DATA FLOW ARCHITECTURE





DIMENSIONAL MODEL / BUSINESS PROCESSES

Fact Table	fact_product_sales
Granularity	One row per order line item order_id + line_no
Dimensions	dim_date: When transaction occurred. dim_product: Product details dim_customer (SCD Type 2): Customer information over time. dim_store: Store/location of purchase. dim_currency: Currency standardisation.

Sales business process records every individual product purchase made by ICE Entertainment customers across Music, Films and Audio Books.

Each transaction line is stored separately enabling granular analysis by product category, store location and customer demographic.



DIMENSIONAL MODEL / BUSINESS PROCESSES

Fact Table	fact_subscription_revenue
Granularity	One row per subscription subscription_id
Dimensions	dim_date: Subscription timeline. dim_package: Subscription plan details. dim_customer (SCD Type 2): Subscriber details. dim_store: Channel/store.

Subscription business process records customer enrolments into ICE Entertainment content packages.

Customers pay a fixed monthly fee for the duration of their subscription providing ICE with a predictable recurring revenue stream tracked alongside product sales.

dim_date
date_key INT
date CHAR(10)
system_date CHAR(10)
sql_date DATE
julian_date INT
day TINYINT
day_of_the_week TINYINT
day_name VARCHAR(9)
day_of_the_year SMALLINT
week_number TINYINT
month TINYINT
month_name VARCHAR(9)
short_month_name CHAR(3)
quarter CHAR(2)
year SMALLINT
fiscal_week TINYINT
fiscal_period CHAR(4)
fiscal_quarter CHAR(3)
fiscal_year CHAR(6)
week_day TINYINT
us_holiday TINYINT
uk_holiday TINYINT
month_end TINYINT
period_end TINYINT
short_day TINYINT
source_system_code TINYINT
create_timestamp DATETIME
update_timestamp DATETIME
Indexes

dim_currency
currency_sk INT
currency_code CHAR(3)
currency_name VARCHAR(50)
effective_date DATE
currency_rate DECIMAL(10,6)
source_system_code TINYINT
create_timestamp DATETIME
update_timestamp DATETIME
Indexes

dim_package
package_sk INT
package_id INT
name VARCHAR(100)
description VARCHAR(255)
package_type CHAR(2)
package_type_desc VARCHAR(50)
package_price DECIMAL(10,2)
source_system_code TINYINT
create_timestamp DATETIME
update_timestamp DATETIME
Indexes

dim_product_type
product_type_sk INT
product_type_code CHAR(2)
product_type VARCHAR(100)
product_category_sk INT
product_category VARCHAR(50)
source_system_code TINYINT
create_timestamp DATETIME
update_timestamp DATETIME
Indexes

fact_subscription_revenue
subscription_id INT
date_key INT
package_sk INT
customer_sk INT
store_sk INT
package_price DECIMAL(12,2)
monthly_revenue_usd DECIMAL(12,2)
status VARCHAR(20)
months_active INT
source_system_code TINYINT
create_timestamp DATETIME
update_timestamp DATETIME
Indexes

dim_product_category
product_category_sk INT
product_category VARCHAR(50)
description VARCHAR(255)
source_system_code TINYINT
create_timestamp DATETIME
update_timestamp DATETIME
Indexes

dim_product
product_sk INT
product_code INT
name VARCHAR(255)
description VARCHAR(255)
title VARCHAR(255)
artist_code VARCHAR(20)
product_type_code CHAR(2)
product_type_sk INT
product_type VARCHAR(100)
product_category_sk INT
product_category VARCHAR(50)
format VARCHAR(20)
unit_price DECIMAL(10,2)
unit_cost DECIMAL(10,2)
status CHAR(2)
source_system_code TINYINT
create_timestamp DATETIME
update_timestamp DATETIME
Indexes

dim_store
store_sk INT
store_number INT
store_name VARCHAR(100)
store_type VARCHAR(30)
city VARCHAR(100)
state VARCHAR(10)
zipcode VARCHAR(20)
country CHAR(2)
region VARCHAR(50)
division VARCHAR(10)
source_system_code TINYINT
create_timestamp DATETIME
update_timestamp DATETIME
Indexes

fact_product_sales
order_id INT
line_no INT
product_sk INT
date_key INT
customer_sk INT
store_sk INT
currency_sk INT
qty INT
price DECIMAL(10,2)
unit_cost DECIMAL(10,2)
dollar_sales_usd DECIMAL(12,2)
cost_usd DECIMAL(12,2)
margin_usd DECIMAL(12,2)
source_system_code TINYINT
create_timestamp DATETIME
update_timestamp DATETIME
Indexes

dim_customer
customer_sk INT
customer_number INT
customer_type CHAR(1)
name VARCHAR(100)
gender CHAR(1)
date_of_birth DATE
age_group VARCHAR(20)
city VARCHAR(100)
state CHAR(3)
zipcode INT
country CHAR(2)
occupation VARCHAR(100)
household_income INT
status CHAR(2)
permission CHAR(1)
preferred_channel1 VARCHAR(255)
preferred_channel2 VARCHAR(255)
interest1 VARCHAR(50)
interest2 VARCHAR(50)
interest3 VARCHAR(50)
market VARCHAR(30)
effective_from DATE
effective_to DATE
is_current_flag CHAR(1)
source_system_code TINYINT
create_timestamp DATETIME
update_timestamp DATETIME
Indexes



DIMENSIONAL MODEL IMPLEMENTATION

1

Physical Database Construction - MySQL dw

→ Database setup

Single MySQL database dw created housing all 18 tables. One database eliminates cross-system joins and ensures a single source of truth for all ICE reporting.

→ Keys and constraints

Every dimension table carries an AUTO_INCREMENT surrogate key. Foreign keys in fact tables enforce referential integrity. NOT NULL and DEFAULT constraints applied throughout.

→ Table structure

8 stg_ tables mirror source data exactly.
8 dim_ tables hold cleaned conformed attributes.
2 fact_ tables store measurable business events.

→ Date dimension

dim_date populated via MySQL stored procedure generating 1,826 rows from 2022 to 2026 with calendar, fiscal and holiday attributes.



DIMENSIONAL MODEL IMPLEMENTATION

2

Source Data Extraction: Pentaho Data Integration

9 Pentaho transformations extract data from Microsoft Excel source files representing the Informix Sales System and Oracle ERP System. Each transformation is dedicated to one target table ensuring clear separation of concerns.

→ Sales system sources

Order_xHeader · Order_Details · Customer · Store ·
Subscription · Package loaded via Excel Input steps

→ ERP system sources

Product · Product_Type · Product_Category ·
Currency_Rate · Currency loaded via Excel Input steps



DIMENSIONAL MODEL IMPLEMENTATION

3 Transform and Load — Dimension and Fact Tables

→ Currency conversion

Calculator step computes $\text{dollar_sales_usd} = \text{qty} \times \text{price} \times \text{currency rate}$ and $\text{margin_usd} = \text{sales} - \text{cost}$. All figures stored in USD eliminating runtime conversion.

→ SCD Type 2 — `dim_customer`

Modified JavaScript sets `effective_from`, `effective_to` and `is_current` flag. Historical market assignments preserved without reprocessing fact data.

→ Age group derivation

JavaScript derives `age_group` from date of birth into three segments: `≤30` · `31–50` · `50+` stored directly in `dim_customer`.

→ Job orchestration

1 Pentaho Job runs all 9 transformations in dependency order. Dimensions load before facts. Truncate and reload ensures clean data.

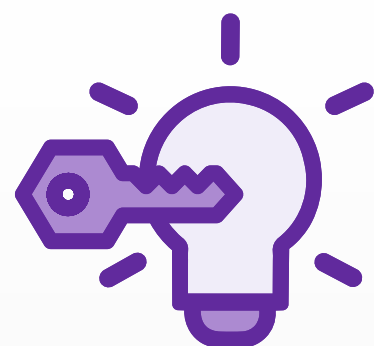


DIMENSIONAL MODEL IMPLEMENTATION

4

Data Verification - Row Counts and Reconciliation

Table	Verification method	Result
dim_date	Stored procedure output checked against date range	1,826 rows ✓
dim_product	Row count matched against source Product.xlsx	1,000 rows ✓
dim_customer	Age groups verified via TIMESTAMPDIFF SQL update	5,000 rows ✓
dim_store	State NULL check for non-Australian stores verified	500 rows ✓
dim_currency	7 currencies × monthly rates reconciled	369 rows ✓
fact_product_sales	Surrogate keys joined back to dimensions and margin calculated	44,443 rows ✓
fact_subscription_revenue	subscription_id, package_sk and customer_sk all populated	504 rows ✓



KEY FINDINGS & BUSINESS INSIGHTS

Q1- Who are the key customers?

market	total_subscription_revenue_usd
International	9878.48
Rest of Australia	785.67
Victoria	480.81

market	total_margin_usd
International	125893.51
Rest of Australia	9082.91
Victoria	6613.36

International Market Dominates Both Revenue Streams

- **87.4% of product margin** • **88.6% of subscription revenue**

International customers are the most strategically valuable segment to ICE Entertainment, generating the overwhelming majority of both product margin and subscription revenue.

Consistent Ranking Across Both Fact Tables

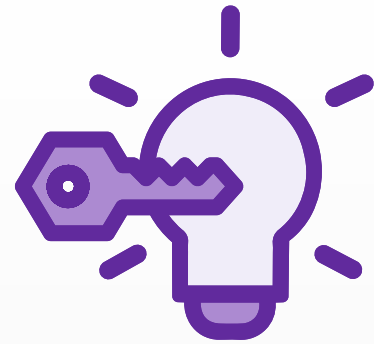
- **International** ► **Rest of Australia** ► **Victoria**

The same hierarchy holds across both one-off product purchases and recurring subscriptions, confirming International leadership is structural rather than channel-specific.

Loyalty Program Should Prioritise International Customers

- **Highest-priority segment** • **ICE Entertainment's first loyalty program**

Their dominance across both revenue streams makes International customers the clear anchor segment for the launch of ICE Entertainment's first customer loyalty program.



KEY FINDINGS & BUSINESS INSIGHTS

Q2-Which products are the most profitable?

product_category	total_margin_usd	margin_pct	total_orders
Audio Books	52427.74	45.16	9871
Films	51384.87	37.24	9763
Music	37777.17	58.91	10148

product_category	product_type	title
Audio Books	Traditional Bk	Code Business
Audio Books	HR Books	Principles Change
Audio Books	Engineering Bk	Wisdom Success
Audio Books	Cooking Books	Way Growth

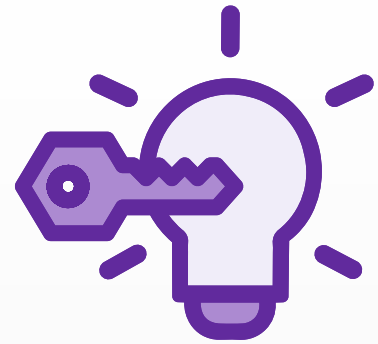
Audio Books Generates Most Absolute Profit

- **\$52,427.74 total margin**
- **45.16% margin percentage**

Audio Books ranks first in total margin USD making it the most profitable category in absolute dollar terms for ICE Entertainment despite having fewer orders than Music.

Traditional book, HR books, Engineering Book, Cooking Book are the most popular product type within audio books

All from Audio books category.



KEY FINDINGS & BUSINESS INSIGHTS

Q3- Which store locations are the most profitable?

store_name	store_type	division	region	margin_pct
Naples20	Distribution Centre	EMEA	Italy	50.01
Liverpool20	Mini Outlet	EMEA	United Kingdom	47.53
Bradford16	Full Outlet	EMEA	United Kingdom	46.50
Leeds18	Mini Outlet	EMEA	United Kingdom	46.23

store_name	store_type	margin_pct	division	region
Perth6	Distribution Centre	48.90	AAA	Australia
Perth12	Full Outlet	47.26	AAA	Australia
Sydney13	Full Outlet	39.17	AAA	Australia
Adelaide10	Full Outlet	44.17	AAA	Australia
Canberra15	Mini Outlet	39.47	AAA	Australia

Amongst all the stores Naples20 is the most profitable

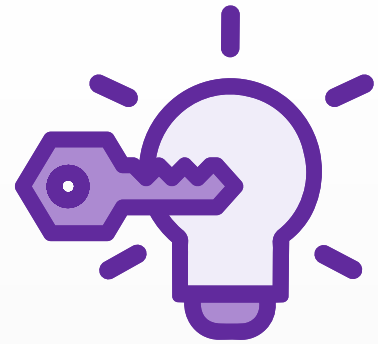
- **50.01% margin percentage**

Overall EMEA division has the highest contributes when it comes to margin percentage

Amongst all the stores in Australia Perth6 is the most profitable

- **48.90% margin percentage**

There are two stores in Perth that makes to top 5 stores in Australia region



KEY FINDINGS & BUSINESS INSIGHTS

Q4- Which time periods are the most profitable?

	calendar_month	dollar_sales_usd	margin_pct
▶	January	32489.14	44.51
	July	26963.07	44.21
	February	26453.88	44.37
	April	25828.63	44.43

	calendar_quarter	fiscal_quarter	dollar_sales_usd	margin_pct	total_orders
▶	Q1	FQ2	60822.75	44.42	2901
	Q4	FQ1	52209.30	44.33	2396
	Q2	FQ3	51556.78	44.49	2404
	Q3	FQ4	50763.27	44.56	2451

January is the most profitable month

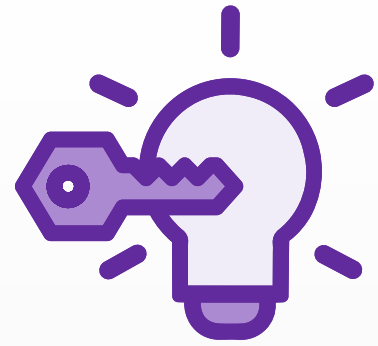
- **44.51% margin percentage**

Overall dollar sales in January is \$32K

Quarter 1 is most profitable

- **44.42% margin percentage**

Total orders in Q1 are significantly high



KEY FINDINGS & BUSINESS INSIGHTS

Q5- Which market is the most profitable?

market	total_customers	total_orders	unit_sales	dollar_sales_usd	cost_usd	margin_usd	margin_pct
International	4227	13426	84737	282929.97	157041.70	125893.51	44.50
Rest of Australia	312	983	6409	20396.98	11315.10	9082.91	44.53
Victoria	216	673	4327	14895.11	8282.21	6613.36	44.40

International market outperforms the other market

- **Highest customer**
- **44.50% margin percentage**
- **Total sales**

Outperforms Rest of Australia and Victoria in all categories



CONCLUSION

With ICE Dimensional Model

- Single dw database answers all 5 business questions
- SCD Type 2 preserves market history without reprocessing
- USD conversion enables fair global comparison across 10 countries

Without Dimensional Model

- Disconnected source systems with no unified customer view
- No historical address tracking or market change management
- Multi-currency data with no common USD comparison

**THANK
YOU**